System Validation Report

Online Exam Portal | Software Engineering Mini-Project, Part 3 | 6 November 2026

Prepared for Chiranthan V | PES2UG24CS138 | Section C | Test lead

Tested PR head: fa38c385bd64c86603fa4970db873971b994eaf8

GitHub executed merge: ee9b05cf0ee801885530630d0f4e6af7c6d3462d. Its file tree equals the PR head (verified with git rev-parse).

Report-only commits may follow; they do not change the application measured in this report. Exact historical operational revisions are stated separately.

1. Validation decision

Automated integrated-product validation PASSED in the named CI environment. Final acceptance/release freeze is PENDING independent SCRUM-22/23 review; live SMTP receipt remains user-deferred. Neither an instructor signature nor unconditional satisfaction of the Test Plan's final exit criteria is asserted.

2. Environment and test-plan alignment

GitHub-hosted Linux, configured Python 3.11 and PostgreSQL 16; React/Vite frontend with Node 24.22.2 and Chromium 156.0.8078.4 in the authenticated audit. Backend tests use a fresh test_portal database. Browser tests seed synthetic development records; recovery tests create unique users/attempts. The Lighthouse job uses its own disposable SQLite data. Container TLS smoke uses a temporary Caddy CA and DEBUG=0 PostgreSQL stack.

Tool substitutions from the pre-execution plan: Django unittest/DRF client for backend testing; Playwright for browser flows; custom benchmark for load; explicit injection payload regressions rather than a claimed ZAP scan. Original PDFs remain unchanged. No real students or new production hosting are used for Part 3.

3. Fresh automated execution

Validation	Outcome	Evidence
Backend functional/security/concurrency	100/100 passed	backend-tests.log; five genuine PostgreSQL row-lock tests
Monitor regression	9/9 passed	Backend CI step + local-monitor-tests.log
Frontend unit/component	16/16 passed	unit-results.json
Real Chromium system flows	10/10 passed; 0 flaky/skipped	e2e-results.json; 22.238 seconds
Authenticated accessibility	15/15 recorded findings passed	accessibility-summary.json; Lighthouse 100 at 1440/390/320px
Migration/lint/build/deploy/TLS	CI passed	Both workflow step records; tls-verification.txt

Browser scenarios cover administrator question editing and XSS-as-text, partial roster import/membership, exam scheduling/manual/random selection/batches, student eligibility/mobile layout, reset/lockout, complete timed exam with offline recovery/focus/contrast, faculty reports/overrides, queue isolation/age, renderer crash plus persistent browser restart, and released results after exam end. No browser scenario failed, retried or was skipped.

4. Updated case design and requirement coverage

The updated Test Case Document contains 41 scenario groups and maps all 35 SRS requirements (35/35 = 100% design traceability). This is not a claim that every manual/operational test or approval has passed. Test Plan section 13 example IDs are preserved; new P3 IDs identify changed/new scenarios. The missing original companion workbook is explicitly disclosed.

Requirement	Updated case(s)	Validation scope / residual gate
OEP-F-001	TC-AUTH-01, TC-AUTH-02, P3-AUTH-03	Automated functional/security regression passed.
OEP-F-002	TC-AUTH-04	Automated functional/security regression passed.
OEP-F-003	TC-AUTHZ-01	Automated functional/security regression passed.
OEP-F-004	P3-RESET-01, P3-MAIL-02	Reset/outbox tests pass; real SMTP receipt deferred.
OEP-F-005	P3-QB-01	Automated functional/security regression passed.
OEP-F-006	P3-QB-02	Automated functional/security regression passed.
OEP-F-007	TC-SEC-01	Automated functional/security regression passed.
OEP-F-008	P3-EXAM-01	Automated functional/security regression passed.
OEP-F-009	P3-EXAM-02	Automated functional/security regression passed.
OEP-F-010	P3-EXAM-03	Automated functional/security regression passed.
OEP-F-011	TC-REG-01, TC-REG-02, TC-REG-03	Automated functional/security regression passed.
OEP-F-012	TC-BATCH-01	Automated functional/security regression passed.
OEP-F-013	P3-EXAM-04	Automated functional/security regression passed.
OEP-F-014	P3-ATTEMPT-01, P3-ATTEMPT-02	Automation passed; independent SCRUM-22/23 acceptance pending.
OEP-F-015	TC-TIMER-02, TC-TIMER-03	Automation passed; independent SCRUM-22/23 acceptance pending.
OEP-F-016	TC-TIMER-01, P3-TIMER-04	Automated functional/security regression passed.
OEP-F-017	TC-RESULT-01, TC-RESULT-02, P3-RESULT-03	Automated functional/security regression passed.
OEP-F-018	TC-SEC-06	Automated functional/security regression passed.
OEP-F-019	P3-REPORT-01, P3-REPORT-02	Automated functional/security regression passed.
OEP-F-020	TC-PROC-01	Automated functional/security regression passed.
OEP-F-021	TC-PROC-02	Automated functional/security regression passed.
OEP-F-022	P3-MAIL-01, P3-MAIL-02	Reset/outbox tests pass; real SMTP receipt deferred.
OEP-NF-001	TC-PERF-01	Dated synthetic benchmark / PostgreSQL evidence; not fresh 200-browser production load.
OEP-NF-002	P3-SCALE-01	Dated synthetic benchmark / PostgreSQL evidence; not fresh 200-browser production load.
OEP-NF-003	P3-UPTIME-01	5 Nov declared controlled-window measurement passed; no production SLA.

Requirement	Updated case(s)	Validation scope / residual gate
OEP-NF-004	TC-TIMER-02, TC-TIMER-03	Automation passed; independent SCRUM-22/23 acceptance pending.
OEP-NF-005	P3-TIMER-04, P3-A11Y-01	Fresh Lighthouse/keyboard checks pass; 5 Nov native-zoom audit scoped to basics.
OEP-NF-006	P3-COV-01	Measured backend core coverage exceeds 70%; exclusions disclosed.
OEP-SR-001	TC-AUTH-01	Automated functional/security regression passed.
OEP-SR-002	P3-AUTH-03, P3-ATTEMPT-02, TC-SEC-04	Fresh CI TLS1.2/cookie/header check passed; temporary test stack.
OEP-SR-003	TC-SEC-01	Automated functional/security regression passed.
OEP-SR-004	TC-AUTHZ-01	Automated functional/security regression passed.
OEP-SR-005	P3-ATTEMPT-01, P3-ATTEMPT-02	Automation passed; independent SCRUM-22/23 acceptance pending.
OEP-SR-006	TC-PROC-02, TC-SEC-05	Automated functional/security regression passed.
OEP-SR-007	TC-REG-02, P3-REPORT-02, TC-SEC-03	Automated functional/security regression passed.

5. Dated operational evidence and defect closure

Availability: 5 Nov source 216fd043df0ff91b3be8c5f6a9f5755138014133, declared 06:03:23-06:33:23 UTC / 11:33:23-12:03:23 IST. External trusted HTTPS probe covered 1,800 seconds; 100.0% slot-weighted estimate, zero failed probes/downtime estimate, 362 healthy guarded raw probes. Two hundred synthetic API identities produced 24,200 origin requests without errors. Tunnel retired. This establishes only the announced controlled test window; loopback load excludes external tunnel/TLS costs and is not 200 full browsers.

Performance: preserved 1 Nov benchmark source/merge ff666788a3fa795e2e19ce537f2bca1743d6b0e8 / 58d7e35d8f83b1ff297f70d388fc5863575ab0a8 reports 200 identities, 3,000 requests, zero unexpected outcomes and five real Chromium openings. Start p95 1.761s; save 0.153s; resume 0.170s; submit 0.285s; page-opening samples 0.123-0.140s. These historical figures are not new measurements of the Part 3 commit or 200 full-browser/WAN capacity.

Accessibility: 5 Nov PR #60 fixed reproduced confirmation-focus loss and sticky-timer focus obscuration. Native Chromium 100/200/400% zoom, keyboard/reflow/forced-colors checks are retained separately. Fresh CI repeats authenticated Lighthouse/keyboard/layout checks, not that entire native-zoom manual session. No full WCAG certification or live screen-reader audio claim.

Expiry: preserved 3 Nov separate-process synthetic worker evidence confirms closure/evaluation/outbox without a browser request. Fresh backend worker tests also pass. Prior broken timer/performance evidence is historical and is not relabelled as success.

6. Remaining acceptance and documented exceptions

- Independent reviewer selected by Chiranthan: Harkirat or Dwijesh. PR #61 checklist is merged and review requests were sent; a merge alone is not independent acceptance. SCRUM-22/23 remain In Review at this checkpoint. No approval attributed in advance.

- SMTP delivery: explicitly deferred by user on 5 Nov after connection timeouts. Outbox and local email tests pass; real controlled-mailbox receipt has not passed. This is not an evaluator-approved waiver.

- Only Chromium/jsdom were executed here. No Safari/Firefox compatibility certification, penetration-test certification or full production exam lifecycle/load/SLA claim.

- No new failing automated assertion or newly reproduced product defect in this Part 3 run. Final acceptance requires genuine peer response, resolution of blockers and final release freeze/tag. Instructor acceptance remains with the evaluator.

Plan exit assessment: automated suite execution is complete for the measured suite with all high-priority automated cases passing; no critical/security failure was observed in those runs. Full planned-case/approval completion cannot be declared while the named deferred/manual acceptance gates remain. Defect density/aging is not fabricated from a non-exhaustive issue history.

Evidence and reproduction

Backend CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42198702219

Frontend/browser CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42198702163

Permanent repository evidence: docs/testing/part3/evidence/2026-11-06/. The manifest records SHA-256 hashes; original backend ZIP was verified against GitHub's artifact digest before extraction. Other original ZIP digests are retained in the manifest; selected raw JSON/text files are preserved without report screenshots.

Detailed coverage: coverage.json, coverage.xml and backend-coverage-original.zip (includes the interactive htmlcov/index.html report). Backend log lists every test; frontend JSON lists each assertion/scenario and actual status.

Related repository records: docs/testing/RTM.md; part3/TEST_CASES.md; availability/RESULT_2026-11-05.md; performance/RESULTS.md; ACCESSIBILITY_CLOSEOUT_2026-11-05.md; closeout/2026-11-03/worker-result.json. Their dated/source-scoped records remain authoritative for those operational results.

```
python -m pip install -r requirements-dev.txt
python backend/manage.py migrate --noinput
python backend/manage.py seed_demo
cd frontend
npm ci
npx playwright install --with-deps chromium
npm run test:e2e -- --reporter=list,html,json
# from repository root on Linux:
bash tools/accessibility/run-ci.sh
```

Use a new synthetic database with DEBUG=1, and seed_demo only in that database. Browser configuration expects .venv/bin/python on Linux, API port 8000 and Vite port 5173; do not reuse unrelated servers. Workflow files contain the complete reproducible PostgreSQL/TLS setup.